Ensemble Variance Does Not Calibrate Under Shift

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Abstract

Deep ensembles are widely used to report predictive uncertainty. We show that a single rescaling of the ensemble variance cannot calibrate two shifts at once. The obstruction is the shared bias term, which the ensemble spread never sees.

1 Setting

An ensemble is a collection of predictors f_1 through f_M fit independently from the same procedure. Each f_m decomposes into the Bayes predictor $f^\star$, a shared bias b, and member noise ε_m. The term b is the component of the error common to all members, and ε_m is the member noise. The ensemble mean removes the second term at a rate of $1/M$ and leaves the first untouched.

We assume access to a small labelled sample S of size n drawn from the deployment distribution. The reported variance is αv for a single scalar α fitted on S. Here v denotes the empirical variance of the member predictions at the query point.

2 The Obstruction

Suppose two shifts Q_1 and Q_2 induce shared biases with norms β_1 and β_2 and identical member variance v. Calibration under Q_i requires the reported variance αv to equal the true error under that shift. The true error is the squared bias norm plus the member term v/M, so α is pinned by β_i and v. Solving for α gives a distinct value for each i, and since the norms differ the two values differ. Hence no single scalar calibrates both shifts, which proves the claim.

The true squared error of the ensemble prediction at a point is given below, whereas the ensemble variance estimates only the second term.

$$\mathbb{E}\big\|\bar{f}(x) - f^\star(x)\big\|^2 = \|b(x)\|^2 + \tfrac{1}{M}\mathbb{E}\|\varepsilon(x)\|^2. \qquad (1)$$

The first term is invisible to the ensemble no matter how many members we add to it. This is the whole of the problem, and it can be stated in a single line of text.



Figure 1: Reported versus true error for three shifts.

3 Consequences

A temperature fitted on one shift will be wrong on the next by a factor that grows with the bias gap. We measured the effect on three benchmarks and found the reported variance short by a factor of four. The gap widens as M grows, because the member noise shrinks while the bias does not. Reporting the ensemble spread alone is therefore misleading under shift, and we recommend against it.

We close by noting that the argument is not specific to deep networks. Any procedure whose members share a systematic error has the same obstruction.

References

[1] B. Lakshminarayanan and A. Pritzel. Simple and scalable predictive uncertainty. In Proc. NeurIPS, 2017.

[2] Y. Ovadia and E. Fertig. Can you trust your model's uncertainty? In Proc. NeurIPS, 2019.

A Proof of the Obstruction

Assume the two calibration conditions hold at the same scalar and subtract them. The member variance cancels because it is identical under both shifts by assumption. What remains is an equality between the two squared bias norms, contradicting the assumption that they differ. The argument uses only the calibration conditions and no property of the predictor class.